The Exact SD-Tuple Boundary Is Reciprocal ℓ^1

Mutually unbiased bases determine the uniform SD-constant

Candidate full result for the SD-tuple question in arXiv:1803.09212

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Status: candidate full proof, likely valid. The packet resolves the precise SD-tuple classification question and computes the uniform SD-constant. It does not solve the source's broader product containment Problem 2.3. Novelty remains subject to expert literature review.

1 Source question and exact scope

Passer defines the standard simplex $\Delta_n = \mathrm{conv}\{0, e_1, \ldots, e_n\}$ and the standard diamond $\lozenge_n = \mathrm{conv}\{\pm e_1, \ldots, \pm e_n\}$, and then introduces SD-tuples on source PDF p. 7 [1]:

constants. These constants are defined in reference to products of simplices and products of diamonds.

Definition 2.4. The *standard simplex* Δ_n in $\mathbb{R}^n$ refers to the convex hull of 0 and the standard basis vectors $e_1, \ldots, e_n$. The corresponding *standard diamond* $\lozenge_n$ is the ℓ^1 unit ball in $\mathbb{R}^n$, i.e., the smallest symmetric convex set containing Δ_n.

Definition 2.5. Call a tuple $(a_1, \ldots, a_d)$ of positive numbers an *SD-tuple* if it holds that for any $n \in \mathbb{Z}^+$,

$$\mathcal{W}^{\max}\left(\Delta_n^d\right) \subseteq \mathcal{W}^{\min}\left(\prod_{j=1}^{d} a_j \cdot \lozenge_n\right). \tag{2.6}$$

Similarly, let

$$\begin{aligned}\mathcal{U}(d) &:= \sup_{n \in \mathbb{Z}^+} \theta(\Delta_n^d, \lozenge_n^d) \\ &= \inf\{C > 0 : (C, \ldots, C) \text{ is an SD-tuple of length } d\}\end{aligned}$$

be called the *uniform SD-constant for d-tuples.*

Dilation techniques from [8] imply that SD-tuples exist and that $\mathcal{U}(d)$ is finite, and such results will be clarified later in this section. The most common use of SD-tuples will come in the following form.

Proposition 2.7. Fix $n_1, \ldots, n_d \in \mathbb{Z}^+$ and let P be a tuple of self-adjoint operators $P_j^{[i]}$, $1 \leq i \leq d$, $1 \leq j \leq n_i$, on $\mathcal{B}(H)$ such that $W_1(P) \subseteq \left(\prod_{i=1}^{d} \Delta_{n_i}\right)$. (That is, $P_j^{[i]} \geq 0$, and

Exact transcription of Definition 2.5. "Call a tuple $(a_1, \ldots, a_d)$ of positive numbers an SD-tuple if it holds that for any $n \in \mathbb{Z}^+$,

$$\mathcal{W}^{\max}(\Delta_n^d) \subseteq \mathcal{W}^{\min}\left(\prod_{j=1}^{d} a_j \cdot \lozenge_n\right).$$

" The uniform SD-constant is

$$\mathcal{U}(d) = \inf\{C > 0 : (C, \ldots, C) \text{ is an SD-tuple of length } d\}.$$

The source proves a reciprocal ℓ^1 sufficient condition and a reciprocal ℓ^2 necessary condition, leaving the gap shown on source PDF p. 11:

> the joint spectrum of N is contained in $L([a_1,\dots,a_d])$, and we may conclude that (2.22) holds. Next, (2.23) follows from (2.22), as any tuple of operators with numerical range in K_i admits a normal dilation with joint spectrum in $\mathcal{W}^{\min}(\theta(K_i)K_i)$, by definition. Finally, (2.24) follows immediately from (2.23). ∎
>
> **Corollary 2.25.** If $(a_1,\dots,a_d)$ is a tuple of positive numbers with $\sum_{i=1}^{d} a_i^{-1} \leq 1$, then $(a_1,\dots,a_d)$ is an SD-tuple. Consequently, $\mathcal{U}(d) \leq d$.
>
> **Proof.** Theorem 2.21 shows that for any $n \in \mathbb{Z}^+$,
> $$\mathcal{W}^{\max}(\Delta_n^d) \subseteq \mathcal{W}^{\min}\left(\prod_{j=1}^{d} \theta(\Delta_n) a_j \Delta_n\right) = \mathcal{W}^{\min}\left(\prod_{j=1}^{d} a_j \Delta_n\right) \subseteq \mathcal{W}^{\min}\left(\prod_{j=1}^{d} a_j \diamondsuit_n\right),$$
> so by definition $(a_1,\dots,a_d)$ is an SD-tuple. ∎
>
> All together, we have that for positive tuples $(a_1,\dots,a_d)$,
> $$\frac{1}{a_1}+\ldots+\frac{1}{a_d} \leq 1 \implies (a_1,\dots,a_d) \text{ is an SD-tuple } \implies \frac{1}{a_1^2}+\ldots+\frac{1}{a_d^2} \leq 1, \tag{2.26}$$
> and in particular
> $$\sqrt{d} \leq \mathcal{U}(d) \leq d. \tag{2.27}$$
> Analogous to the computation (2.1), we suspect that if one of the implications is an equivalence, then SD-tuples may be characterized by the identity $\sum_{j=1}^{d} a_j^{-2} \leq 1$. However, the estimates of Theorem 2.21 are optimal for certain positive sets, such as $[0,1]^d$, once again emphasizing that dilation problems concerning symmetric sets are (or have the potential to be) considerably more flexible than those concerning positive sets. Recall that the difficulties of positive-to-positive dilation were abstracted in [21] to *simplex-pointed* sets.

Exact transcription of the question. "All together, we have that for positive tuples $(a_1,\dots,a_d)$,
$$\frac{1}{a_1}+\cdots+\frac{1}{a_d} \leq 1 \Longrightarrow (a_1,\dots,a_d) \text{ is an SD-tuple} \Longrightarrow \frac{1}{a_1^2}+\cdots+\frac{1}{a_d^2} \leq 1,$$
and in particular $\sqrt{d} \leq \mathcal{U}(d) \leq d$. Analogous to the computation (2.1), we suspect that if one of the implications is an equivalence, then SD-tuples may be characterized by the identity $\sum_{j=1}^{d} a_j^{-2} \leq 1$."

We prove that the first implication is an equivalence. Thus the sufficient condition in Corollary 2.25 is sharp, $\mathcal{U}(d) = d$, and the proposed reciprocal ℓ^2 characterization is false for every $d \geq 2$.

2 Two elementary matrix-convex lemmas

For a finite polytope $K = \operatorname{conv}\{v_s : s \in S\} \subset \mathbb{R}^N$, the level-$m$ points of its minimal matrix convex set are precisely
$$X = \sum_{s\in S} v_s F_s, \qquad F_s \succeq 0, \qquad \sum_{s\in S} F_s = I_m. \tag{1}$$
Indeed, this is the matrix convex hull of the scalar vertices; allowing all scalar points of K gives nothing more after splitting their ordinary convex combinations among the effects.

Lemma 1 (Product-diamond coarse graining)**.** *Let $P^{(i)} = (P_1^{(i)},\dots,P_p^{(i)})$ be rank-one projection-valued measures on $\mathbb{C}^p$, for $1 \leq i \leq d$, and put $t_i = a_i^{-1}$. If the conjoined tuple $(P_j^{(i)})_{i,j}$ lies in*
$$\mathcal{W}_p^{\min}\left(\prod_{i=1}^{d} a_i \diamondsuit_p\right),$$

then there are jointly measurable POVMs $E^{(i)} = (E_1^{(i)}, \dots, E_p^{(i)})$ satisfying

$$E_j^{(i)} \succeq t_i P_j^{(i)} \qquad (1 \le i \le d,\ 1 \le j \le p). \tag{2}$$

Proof. The vertices of the product polytope are indexed by $v = ((\varepsilon_1, j_1), \dots, (\varepsilon_d, j_d))$, with $\varepsilon_i \in \{+,-\}$ and $1 \le j_i \le p$; its ith block is $a_i \varepsilon_i e_{j_i}$. Let $(F_v)_v$ be the POVM in (1). For each i, j, sum the effects with ith signed outcome $(+, j)$ or $(-, j)$ to get $H_{j,+}^{(i)}$ and $H_{j,-}^{(i)}$. The coordinate identity is

$$P_j^{(i)} = a_i\big(H_{j,+}^{(i)} - H_{j,-}^{(i)}\big).$$

Consequently $E_j^{(i)} := H_{j,+}^{(i)} + H_{j,-}^{(i)} \succeq t_i P_j^{(i)}$. Summing F_v over all sign choices while retaining $(j_1, \dots, j_d)$ gives a joint POVM for the $E^{(i)}$. □

Lemma 2 (Trace incompatibility witness)**.** *Under the conclusion of Lemma 1,*

$$\sum_{i=1}^{d} t_i \le \max_{j_1,\dots,j_d} \left\| \sum_{i=1}^{d} P_{j_i}^{(i)} \right\|. \tag{3}$$

Proof. Let $G_{j_1,\dots,j_d}$ be a joint POVM and write $J = (j_1, \dots, j_d)$. Since each $P_j^{(i)}$ has trace one, (2) gives

$$p \sum_i t_i \le \sum_{i,j} \operatorname{tr}(P_j^{(i)} E_j^{(i)}) = \sum_J \operatorname{tr}\left[\left(\sum_i P_{j_i}^{(i)}\right) G_J\right].$$

For positive matrices A, G, $\operatorname{tr}(AG) \le \|A\| \operatorname{tr}(G)$. The last display is therefore at most the right side of (3) times $\sum_J \operatorname{tr}(G_J) = \operatorname{tr}(I_p) = p$. Divide by p. □

Proof Intuition

The signed diamond coordinates initially look more flexible than ordinary measurement compatibility. Lemma 1 shows that signs only add positive slack: an SD representation would still produce compatible noisy versions of any chosen sharp measurements, with sharp component $a_i^{-1} P^{(i)}$. To rule this out, choose sharp measurements that become almost disjoint as the matrix size grows. Mutually unbiased bases do exactly that. Their pairwise overlaps are $p^{-1/2}$, so every transversal sum of one projection from each basis has norm $1 + o(1)$. The trace witness then forces the total sharp weight $\sum_i a_i^{-1}$ to be at most one.

3 Exact classification

Theorem 3 (Exact SD-tuple criterion)**.** *For every positive tuple $(a_1, \dots, a_d)$,*

$$(a_1, \dots, a_d) \text{ is an SD-tuple} \iff \sum_{i=1}^{d} \frac{1}{a_i} \le 1. \tag{4}$$

Consequently $\mathcal{U}(d) = d$.

Proof. The reverse implication is Corollary 2.25 of the source [1]. We prove necessity.

Suppose $(a_1, \dots, a_d)$ is an SD-tuple and put $t_i = a_i^{-1}$. Choose an arbitrarily large odd prime p with $p + 1 \ge d$. We first construct d mutually unbiased orthonormal bases of $\mathbb{C}^p$. Let $\zeta = e^{2\pi i/p}$. Along with the standard basis, use the bases

$$u_{r,k}(x) = p^{-1/2} \zeta^{rx^2 + kx}, \qquad x, k \in \mathbb{F}_p, \qquad r = 0, \dots, d-2. \tag{5}$$

Within each fixed r, character orthogonality makes these vectors an orthonormal basis. The standard-basis overlaps have modulus $p^{-1/2}$. For $r \neq s$, the remaining assertion follows from the elementary quadratic Gauss sum

$$\left| \sum_{x \in \mathbb{F}_p} \zeta^{Ax^2+Bx} \right|^2 = p \qquad (A \neq 0). \tag{6}$$

For completeness, expand the square and put $x = y + h$. When $h = 0$ the inner sum over y is p; when $h \neq 0$, its coefficient $2Ah$ is nonzero and the inner character sum vanishes. This proves (6).

Let $P_j^{(i)}$ be the rank-one projections associated with these bases. Each block is a PVM, so it belongs to $\mathcal{W}_p^{\min}(\Delta_p)$: use effects $P_1^{(i)}, \dots, P_p^{(i)}$ at the nonzero simplex vertices and the zero effect at the zero vertex. Since the maximal matrix convex set over a Cartesian product is the product of the maximal sets, the conjoined tuple is in $\mathcal{W}_p^{\max}(\Delta_p^d)$. The SD hypothesis and Lemma 2 therefore give

$$\sum_i t_i \leq \max_{j_1,\dots,j_d} \left\| \sum_i P_{j_i}^{(i)} \right\|. \tag{7}$$

Fix a transversal and let $U : \mathbb{C}^d \to \mathbb{C}^p$ have the selected unit vectors as columns. The sum of projections is UU^*, while its Gram matrix is U^*U; their nonzero eigenvalues agree. The Gram matrix has diagonal entries one and every off-diagonal entry has modulus $p^{-1/2}$. Gershgorin's theorem thus yields

$$\left\| \sum_i P_{j_i}^{(i)} \right\| = \|U^*U\| \leq 1 + \frac{d-1}{\sqrt{p}}. \tag{8}$$

Combining (7)–(8) and letting p range through arbitrarily large primes gives $\sum_i t_i \leq 1$, proving (4).

Finally, $(C, \dots, C)$ satisfies (4) exactly when $d/C \leq 1$, so the infimum in the definition of $\mathcal{U}(d)$ is d. □

Corollary 4 (Explicit refutation of the proposed boundary)**.** *For every $d \geq 2$, $(\sqrt{d}, \dots, \sqrt{d})$ satisfies $\sum_i a_i^{-2} = 1$ but is not an SD-tuple.*

Proof. Its reciprocal ℓ^1 sum is $\sqrt{d} > 1$, so Theorem 3 applies. □

The obstruction is finite, not merely asymptotic. If $s = \sum_i a_i^{-1} > 1$, choose an odd prime $p \geq d-1$ such that

$$\frac{d-1}{\sqrt{p}} < s - 1. \tag{9}$$

Then the MUB tuple above is an explicit witness that the defining inclusion fails already for simplex dimension $n = p$ at matrix level p.

4 Verification, limitations, and novelty

The accompanying verifier constructs the bases (5), checks orthonormality and mutual unbiasedness, forms transversal projection sums, and compares their norms with (8). Exhaustive enumeration was performed for $(p, d) = (3, 2), (5, 3), (7, 4), (11, 5)$ (up to 161,051 transversals), together with 5,000 sampled transversals for $(17, 7)$. Every check passed, with orthogonality and unbiasedness errors below $6 \cdot 10^{-16}$. These computations are sanity checks; the proof is analytic and self-contained apart from the source's already-proved sufficient direction.

Adversarial checks. The proof uses the all-n quantifier in the SD definition, not a fixed low matrix level. It does not assume that signed marginals are positive differences: both signed marginal effects are positive by construction, so their sum dominates their difference. The trace inequality remains valid without commutativity because both matrices are positive. Collisions among joint outcomes are handled by coarse graining the original product-vertex POVM.

Scope limitation. This is a full answer to the precise classification of SD-tuples and the value of $\mathcal{U}(d)$. Problem 2.3 in the source asks a much broader question about arbitrary compact convex factors and target sets; that general program is not resolved here.

Bounded novelty audit. The run indexes and primary-source arXiv searches through 13 August 2026 were checked using arXiv:1803.09212 and the phrases "SD-tuple," "uniform SD-constant," and "minimal matrix convex product." The search found the source and later papers citing it, but no statement of criterion (4) or $\mathcal{U}(d) = d$. Novelty is plausible, not certified.

Human-review recommendation: send to a matrix-convexity/operator theory expert. The highest-value review points are the polytope POVM representation (1), the signed-to-unsigned coarse graining in Lemma 1, and the placement of the MUB PVM tuple in $\mathcal{W}_p^{\max}(\Delta_p^d)$.

References

[1] B. Passer, *Shape, Scale, and Minimality of Matrix Ranges*, Trans. Amer. Math. Soc. **372** (2019), 1451–1484; arXiv:1803.09212.